

NBA Player Archetype Evolution

Milestone 3

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Analytical Question

The focus of this project is to assess if the traditional NBA position titles still define the functions and actions of players in the game today or if there was a true evolution of player archetypes since the onset of the analytics movement in the league around 2015. The primary question is can data clustering separately define the new position groupings that were not possible previously today because five original positions cannot accurately define how players play anymore? Since the ability to shoot the three-point shot was added to the game in 1979, and before the analytics movement became a integral to the NBA around 2015, player roles and functions have blurred to where there are very similar statistical profiles for both a Center and Point Guard. We are going to take both per-game and advanced metrics data for 1980-2026, with player height comparisons as a constant to create new defined groupings to demonstrate how positions have defined and evolved over time.

Data & Methodology

This dataset was created via Kaggle, and the original data originates from Basketball Reference. This source documents all seasons for all NBA players 1980-2026. Filters were applied to only include games played by players that participated in 40+. To be included for average per game metrics, players had to average 20+ minutes in their 40+ games played, yielding 6,243 player season entries with complete height info. To compare each group of players by era, we divided them into 3 eras of pre-2000, 2000-2014, and 2015-present. This complete pipeline process was constructed utilizing Python (Pandas), Matplotlib, Seaborn, Plotly, and Streamlit.

Key Findings

- **The 3-Point Revolution:** Average three-point attempts per game rose from 1.7 in 2000 to 4.4+ by 2026, with the sharpest acceleration after 2015. Centers increased their 3PA by +794% and Power Forwards by +134% since 2015 alone.
- **Height of Perimeter Shooters Rising:** The average height of players attempting 2+ threes per game climbed from 6'5" in 1990 to 6'7" by 2026, showing that taller players are driving the perimeter revolution — not just guards.
- **The Efficiency Inversion:** In the modern era, Centers now lead all positions in Win Shares Per 48 minutes at 0.15 — the highest of any position in any era — while guard efficiency has declined. Expanded skill sets have made tall players the most efficient per minute.
- **Playmaking and Rebounding Converging:** The 6'9"+ group increased assists by +27% since pre-2000 while maintaining the highest rebounding numbers of any height group, showing that two historically separate skill sets are now merging into single players.

Limitations

- **Listed positions are already biased:** Teams assign position labels that are themselves outdated, meaning the variable used for comparison already contains the problem this project is trying to expose.
- **Minimum filters exclude role players:** The 40-game, 20-minute threshold creates survivorship bias toward high-usage players and may not fully represent all player types.
- **Stats do not capture everything:** Some of the most valuable players in NBA history have been those whose contributions are least visible in box scores — screen setting, defensive communication, floor spacing without the ball, and locker room leadership are entirely absent from this dataset. A player's true impact on winning cannot always be measured.
- **Era comparisons are imperfect:** Rule changes across decades such as hand-checking rules and pace adjustments inflate or deflate certain stats at different points in history, making direct numerical comparisons across eras less precise.